Power-Line Fault Detection

Conversation archive · Low-voltage pulse reflection and sensors

Nata and ChatGPT

About this archive

This archive contains the user questions and substantive assistant replies visible in the current conversation, in their original sequence, through the repeated request to download the chat. Repeated assistant answers are retained. It is not an export of other conversations or of the user's account history.

Archive conventions. Formatting is adapted for offline reading. Mathematical expressions are typeset; the original branch illustration is reproduced as a static figure; interactive controls are represented by their visible text. User spelling is preserved. System instructions, private reasoning, tool-operation messages, and interface timing notices are excluded. The preparatory messages for producing this archive are also excluded.

Sources and technical status. Source links visible in the conversation are retained; they have not been re-verified for this export. The first answer contains chat citation references for which no underlying source URLs are available in the export context. Those are marked as unavailable rather than replaced with invented references. Earlier technical statements are preserved as conversation history, not newly endorsed; later replies include qualifications about pulse width, resolution, and safety.

Existing research files. The complete-download ZIP includes the previously generated paper in PDF and Word formats and the earlier research-package ZIP. These files are included unchanged. Their contents are not duplicated in the chat transcript; the earlier message announcing the paper is included below.

Safety reminder from the discussion. The proposed laboratory baseline is an isolated, de-energized, verified-dead line. The transcript is not a validated construction manual or authorization to connect experimental equipment to energized 380 V conductors.

Conversation contents

1. Pulse-width question and initial answer.
2. Questions about reflection-measurement methods, sensors, and a master's-level paper.
3. Research-paper delivery message.
4. Request to answer the second question, followed by the two assistant replies visible in this conversation.
5. The two requests to download the chat.

01 · User — Pulse-width question

what is the best value of pulse width for measuring failure point (open and short) of 3P 380V parrallel power line by low voltage pulse reflection method?

02 · Assistant — Initial pulse-width answer

For detecting **open-circuit and short-circuit faults in a 3-phase, 380 V power line using Low-Voltage Pulse Reflection (LVPR), also called Time-Domain Reflectometry (TDR)**, I recommend starting with a **pulse width of 20–50 ns** for short cable runs.

However, the optimal pulse width depends primarily on the **cable length, propagation velocity, attenuation, and required fault-location resolution**, rather than the 380 V rating itself.

1. Recommended pulse widths

The following are practical initial settings, not universal optimal values.

Cable length	Suggested pulse width	Application
0–20 m	2–10 ns	Very close faults, high resolution
20–100 m	10–20 ns	Short industrial power cables
100–300 m	20–50 ns	Typical short-to-medium runs
300–1,000 m	50–200 ns	Longer power cables
1–3 km	200–500 ns	Long-distance fault detection
Over 3 km	500 ns–2 µs	Attenuated reflections

The reasoning is straightforward: **shorter pulses improve the ability to distinguish nearby reflections, whereas longer pulses can improve detection on attenuating cables**. This is why TDR instruments commonly provide selectable pulse widths. [Original chat citation: underlying source URLs unavailable in the export context.]

For an industrial 380 V distribution cable approximately 100–300 m long, I would initially test with a 20 ns pulse, then compare the results at 50 ns and 100 ns.

2. Mathematical relationship between pulse width and resolution

For a rectangular pulse, a useful approximation of the spatial pulse length (expressed as one-way distance resolution) is:

$$\Delta x \approx \frac{vT_p}{2}$$

Where:

- Δx: approximate ability to separate reflections, in meters
- v: pulse propagation speed, in m/s
- T_p: pulse width, in seconds

Assume a propagation velocity of:

$$v = 1.7 \times 10^8 \text{ m/s}$$

This is an illustrative value; the actual velocity needs to be calibrated for your cable.

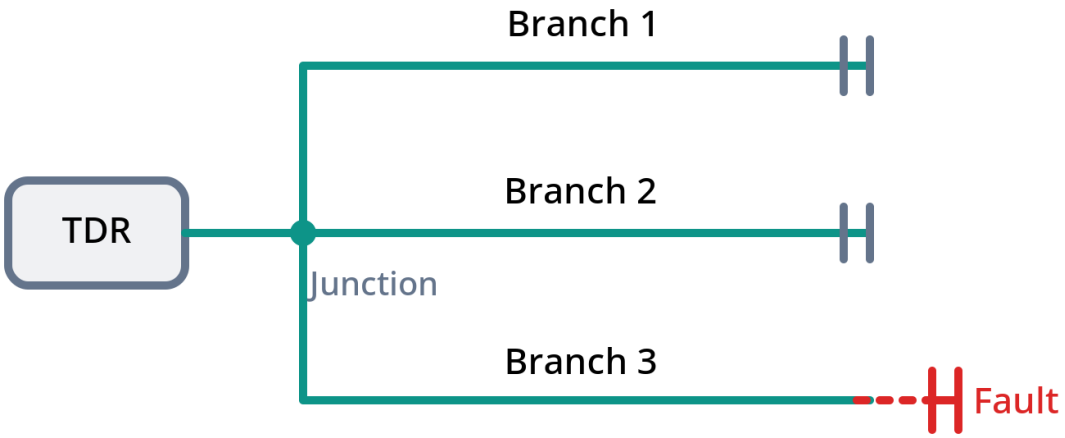
Pulse width	Approximate spatial resolution
5 ns	0.43 m
10 ns	0.85 m
20 ns	1.7 m
50 ns	4.25 m
100 ns	8.5 m
200 ns	17 m
500 ns	42.5 m
1 µs	85 m

These values describe nominal pulse-width-related separation, **not guaranteed fault-location accuracy**. Timing precision, pulse rise time, cable dispersion, sampling rate, and signal processing also determine performance. [Original chat citation: underlying source URLs unavailable in the export context.]

3. How parallel power lines affect the measurement

If by *parallel power line* you mean multiple feeders or branches connected in parallel, choosing the correct pulse width is not enough.

Example: Three parallel feeder branches



Conceptual example of a fault in one parallel branch.

At each junction, the characteristic impedance changes, producing a reflection. The pulse then propagates through multiple branches, generating overlapping echoes.

Consequently:

- A short pulse improves the ability to distinguish closely spaced reflections.
- The same fault distance may correspond to different branches.
- Multiple reflections can resemble additional faults.
- Testing branches separately, where practical, makes interpretation more reliable.

If you mean the three phase conductors L1, L2, and L3 running alongside each other rather than branched feeders, the appropriate measurement arrangement is different: each conductor pair must be assessed with its return path and coupling considered.

4. Open versus short circuit

The reflection coefficient is:

$$\Gamma = \frac{Z_f - Z_0}{Z_f + Z_0}$$

Fault	Reflection coefficient	Reflection
Ideal open circuit	+1	Positive
Ideal short circuit	−1	Negative

Both faults can be located using the same pulse width. The reflection polarity is what principally distinguishes them. [Original chat citation: underlying source URLs unavailable in the export context.]

Important safety condition: These recommendations are for a de-energized, isolated, verified-dead cable. Never connect a conventional low-voltage TDR pulse generator directly to an energized 380 V three-phase line. Also isolate connected equipment, discharge stored energy, and follow the appropriate electrical safety procedures.

5. Select a pulse width for your system

TDR pulse-width calculator

Cable length (m)

[Input value unavailable]

Propagation velocity (m/µs)

[Input value unavailable]

Pulse width **ns**

Slider

Current value unavailable.

Range: 5 to 200

Step: 5

Nominal pulse resolution

m

End reflection delay

This idealized calculator does not predict signal attenuation, instrument dead zone, or accuracy in branched cables.

Two important questions

To recommend a more precise pulse width for your 380 V system, I need to understand:

1. **What is the maximum cable length?** For example, 100 m, 300 m, 1 km, or 5 km?
2. **What does “parallel” mean in your system?** Three phase conductors running alongside each other, or multiple cables/feeders connected electrically in parallel?

With those details, I can help determine the pulse width, rise time, sampling frequency, and measurement circuit needed to distinguish open and short faults accurately.

03 · User — Reflection-measurement methods and sensors

explain about what kind of methods are there for measuring reflection pulse of power line and what kinds of sensors are the best for that?

04 · User — Master's-level paper request

can you make master theory paper on the failure detection method based on low voltage pulse reflection method (radar method) of 3P 380V parallel power line?

05 · Assistant — Research-paper delivery

Nata, I've prepared a **34-page master's-level theoretical paper**:

Fault Detection and Localization in Three-Phase 380 V Parallel Power Lines

Low-Voltage Pulse Reflectometry (Radar Method): Theory, Sensing Architecture, and Reproducible Simulation

Download the paper

File	Contents
Paper — PDF	Formatted paper for reading, printing, and offline use.
Editable paper — Word	Editable text, tables, and mathematical equations.
Complete research package — ZIP	Both documents, Python simulation code, numerical results, figures, manuscript source, and reproduction instructions.

What the paper covers

Area	Included treatment
Transmission-line theory	Telegrapher's equations, characteristic impedance, propagation velocity, attenuation, multiconductor modeling, and modal reflections.
Open- and short-circuit detection	Reflection polarity and amplitude; terminal faults versus faults along a continuing line; series and shunt fault resistance.
Parallel-line configurations	Separate analysis of adjacent phase conductors, branched feeders, and feeders connected in parallel at both ends.
Reflection-measurement methods	Pulse and step TDR, baseline subtraction, spread-spectrum methods, multicarrier reflectometry, and frequency-domain approaches.
Sensors and acquisition	Voltage sensing, differential probes, reflection bridges, high-frequency current transformers, Rogowski coils, capacitive pickups, and shunts.
Pulse and receiver design	Pulse duration, rise time, bandwidth, sampling rate, signal-to-noise ratio, pulse overlap, and measurement blind zones.
Fault-location algorithms	Calibration, incident/reflected-wave separation, signed matched filtering, threshold selection, and topology-aware interpretation.
Validation and uncertainty	An experimental validation plan, error budget, propagation-speed uncertainty, limitations, and electrical safety.

The document contains **38 displayed equations, six original simulation figures, and 16 references**. Its reproducible Python study evaluates **12,000 synthetic noisy traces**, examining how pulse width affects fault detection and timing under explicitly stated assumptions.

A central distinction throughout the paper is that **detecting a reflection, identifying its cause, and locating the physical fault are separate tasks**. The analysis therefore treats pulse overlap, timing precision, absolute distance accuracy, and branch ambiguity separately rather than assigning one “best” pulse width to every 380 V line.

Important scope

This is a **theoretical research draft with synthetic simulations**, not a report claiming measurements from your installation. It develops multiconductor theory, while the supplied simulations use clearly identified simplified transmission-line models—not a fully validated three-phase electromagnetic model.

The proposed laboratory baseline is an **isolated, verified-de-energized line**. Energized-line measurement is discussed as a separate, safety-critical application; the paper does not present a homemade interface for direct connection to live 380 V conductors.

Installation-specific pulse widths, sensor choices, and achievable accuracy remain provisional until the **cable length, construction, return path, and electrical connection topology** are established.

06 · User — Request for the second answer

what is the answer of the second question?

07 · Assistant — Methods and sensors, first visible reply

You're right, Nata—I provided the paper but did not directly answer your earlier question:

> **"What methods can measure the reflected pulse of a power line, and which sensors are best?"**

For your **three-phase 380 V line**, my recommended starting point is **wideband voltage measurement with a pulse generator and an oscilloscope**. For a more capable instrument, I would add a **calibrated reflection bridge** to help separate the returning reflection from the outgoing pulse. A **high-frequency current transformer (HFCT)** is useful as an additional measurement channel, but it is not automatically better than voltage sensing. These recommendations build on the conventional voltage-based TDR approach and the use of directional bridges for reflection measurements. (keysight.com)

Safety: The discussion below assumes an isolated, verified-de-energized cable. A qualified person should establish the safe test condition, including isolation from backfeeds and stored energy. A homemade pulse generator, ordinary oscilloscope, or low-voltage differential probe must not be connected directly to an energized 380 V circuit. (osha.gov)

1. What do you actually measure?

You inject an electrical pulse and record the returning waveform. Its **delay** indicates distance, while its **polarity and amplitude** provide information about the impedance change. For a simple, single-mode line:

$$d = \frac{v_p \Delta t}{2}$$

Here, d is the distance to the reflecting point, v_p is the calibrated propagation velocity, and Δt is the round-trip delay after accounting for the test leads. An ideal open circuit produces a positive **voltage** reflection; an ideal short produces a negative one. (keysight.com)

The important distinction is that **the sensor detects a waveform—not "an open circuit" or "a short circuit" directly**. Your analysis must distinguish fault reflections from reflections caused by connectors, branches, and normal terminations. (tek.com)

2. Methods for measuring the reflected pulse

A. Direct voltage measurement — the simplest starting method

A fast voltage receiver or oscilloscope measures the voltage at the cable's input port:

$$V_{\text{measured}}(t) = V_{\text{incident}}(t) + V_{\text{reflected}}(t)$$

This is the basic TDR arrangement. The receiver sees both the outgoing pulse and returning reflections. It works especially well when the important reflection is sufficiently separated from the launch event. (tek.com)

My assessment: Start here for your laboratory prototype. It gives you the actual waveform and is comparatively easy to debug. Near-end faults are more difficult because their echoes may overlap the launch waveform; the source, fixture, and receiver response must therefore be characterized. (keysight.com)

B. Reflection bridge or directional coupler — better separation

A reflection bridge creates a measurement output that preferentially responds to the returning wave while suppressing the outgoing wave. Its ability to distinguish the two directions is called **directivity**. Commercial RF bridges explicitly specify directivity, coupling loss, and operating bandwidth. (keysight.com)

My assessment: This is the architecture I would investigate for your developed fault detector, particularly when launch leakage limits detection of small or early reflections.

However, select the bridge for the **whole pulse spectrum**, not just a high maximum frequency. Its reference impedance, low-frequency response, and insertion loss matter. Also, use a receiver that preserves the **signed waveform**: a power-only detector cannot retain the positive/negative voltage information needed for straightforward open-versus-short interpretation. (keysight.com)

C. Simultaneous voltage and current measurement — an advanced option

For an approximately lossless, single-mode transmission line, define current as positive **toward the cable**:

$$V = V^+ + V^-, \qquad I = \frac{V^+ - V^-}{Z_0}$$

Solving these two equations gives:

$$V^- = \frac{V - Z_0 I}{2}$$

Thus, the reflected voltage can be reconstructed from simultaneous voltage and current measurements.

This is attractive when incident and reflected signals overlap, but it requires accurate channel gain, timing alignment, and an appropriate characteristic impedance Z_0. **For a coupled three-phase line, the scalar equation is only an approximation for a selected propagation mode—not a complete multiconductor model.** Differential measurements and channel deskew are important when characterizing coupled conductors. (tek.com)

My assessment: A useful research extension, but not the easiest first implementation.

D. Reference subtraction and correlation — processing methods

These methods improve interpretation rather than replace the sensor.

Reference subtraction compares the measured trace with a healthy trace acquired under the same configuration. It can highlight changes, although ordinary connection changes can also create differences. TDR comparison and calibration techniques are established approaches. (tek.com)

Correlation reflectometry, including sequence TDR and spread-spectrum TDR, injects a known coded signal and searches for delayed copies in the received signal. These methods are useful in noisy environments and have been developed for monitoring energized systems with appropriate interfaces. (profiles.faculty.utah.edu)

For completeness, **frequency-domain reflectometry** measures reflection versus frequency and transforms the result into a distance or time response. It is an alternative to directly recording a short pulse. (keysight.com)

3. Which sensors are best?

Best primary choice: a wideband voltage-sensing receiver

For your **offline low-voltage pulse-reflection experiment**, I would prioritize a voltage receiver with **low noise, adequate bandwidth, low loading, and a calibrated connection fixture**.

In an integrated instrument, this can be a properly terminated receiver connected to the reflection bridge. In a laboratory setup, a suitable voltage probe may monitor a defined test node. These are different arrangements: **do not attach an arbitrary probe or a 50 Ω oscilloscope input across the cable without accounting for how it changes the measurement circuit.** Fixtures and probe connections can themselves degrade the edge and introduce reflections. (tek.com)

A differential voltage probe is appropriate when the measurement node must be sensed differentially, but **"differential" does not automatically mean electrically isolated or safe for mains connection**. Safety must be established for the complete measurement arrangement. (osha.gov)

Best additional current sensor: an HFCT or wideband pulse-current transformer

An HFCT measures the magnetic field associated with the pulse current and produces a corresponding output voltage. Suitable pulse-current monitors can resolve nanosecond-scale transitions, but performance varies substantially between models. (pearsonelectronics.com)

For your application, evaluate:

Specification	Why it matters
Usable rise time and bandwidth	Determines whether the sensor preserves the reflected pulse edge.
Sensitivity and receiver noise	Determines whether small reflected currents are detectable.
Pulse droop	Indicates how much the output changes during a pulse.
Current-time product and saturation	Limits the pulse magnitude and duration that can be measured without distortion.
Transfer-function calibration	Needed for reliable amplitude interpretation and voltage/current wave separation.

Pearson's selection guidance specifically discusses sensitivity, usable rise time, droop, frequency limits, and saturation constraints. **Choose from those specifications—not merely the maximum current rating.** (pearsonelectronics.com)

One important consequence of magnetic sensing: a clamp enclosing both the outgoing conductor and its equal-and-opposite return measures their **net current**. Those currents can cancel, so clamping around the complete cable may not reveal the differential TDR signal you intended to measure.

Rogowski coil: useful, but not my first choice here

A Rogowski coil produces a signal proportional to the rate of change of current:

$$V_{\text{coil}} \propto \frac{dI}{dt}$$

An integrator is normally used to recover a current-proportional waveform. Its air core avoids magnetic saturation, but the complete coil–cable–integrator system determines bandwidth and noise performance. (pemuk.com)

My assessment: A Rogowski coil is worth considering for large-current transient measurements. For small low-voltage TDR echoes, I would first compare a voltage receiver and a sensitive HFCT. A Rogowski coil can work, but its suitability must be demonstrated at your expected echo amplitude and edge speed—not assumed from its current rating. (pemuk.com)

4. What bandwidth should the sensor have?

Select bandwidth mainly from the **pulse rise time**, not just its width.

For a simple approximately one-pole response:

$$t_{r,\,\text{receiver}} \approx \frac{0.35}{B}$$

The coefficient depends on the receiver's frequency response; Tektronix notes that some oscilloscopes are better described by a coefficient near 0.45. (tek.com)

For example, **assuming a 5 ns source rise time**, the calculation $0.35/5\,\text{ns}$ gives **70 MHz**. However, a receiver with a 5 ns rise time would noticeably broaden that edge; 70 MHz is therefore not a generous design target.

For an initial bench study using roughly **20–100 ns pulses with 5 ns edges**, my provisional choice would be a **150–200 MHz measurement chain and approximately 1 GS/s acquisition**, followed by measurements of actual edge distortion and noise. This is an engineering starting point, not a guaranteed accuracy specification: the source, sensor, fixture, receiver, and cable all contribute to the observed response. (keysight.com)

5. Recommendation for your three-phase parallel line

I would develop your project in this order:

First prototype: Direct voltage-based pulse TDR on an isolated cable specimen, with known open and short terminations.

Improved instrument: A calibrated reflection bridge feeding a low-noise voltage receiver, with healthy-trace comparison.

Research extension: A synchronized HFCT channel for voltage/current wave separation and investigation of coupled-conductor behavior.

For adjacent phase conductors, define the injection pair, return path, and unused-conductor terminations consistently. For electrically parallel feeders or branches, a single trace can contain echoes from several paths; **a better sensor alone cannot identify which branch produced an otherwise indistinguishable echo**. Separate branch tests or additional measurement locations may be necessary. (tek.com)

Bottom line: for your offline low-voltage radar-method detector, I would choose voltage sensing as the primary measurement, a reflection bridge when directional separation is needed, and an HFCT as an optional complementary sensor—not an ordinary mains-current sensor.

08 · Assistant — Methods and sensors, second visible reply

You're right, Nata—I didn't answer your second question directly:

> **What methods can measure the reflected pulse in a power line, and which sensors are best?**

For your low-voltage pulse-reflection project, my preferred starting point is a wideband voltage-measurement circuit and a digital oscilloscope or fast digitizer. A calibrated reflection bridge can help separate the returning echo from the outgoing pulse. A high-frequency current transformer is useful as an additional measurement channel, but it is not essential for a basic TDR system. Voltage-based TDR and directional wave sampling are established measurement approaches. (Keysight)

Safety boundary: The laboratory arrangements below are for an **isolated, de-energized, verified-dead line**, with stored energy discharged and all possible supplies or backfeeds accounted for. Do not connect an ordinary oscilloscope, pulse generator, or experimental TDR bridge directly to live 380 V conductors. Energized testing requires purpose-designed equipment and qualified electrical personnel. (OSHA)

1. Methods for measuring the reflected pulse

It helps to distinguish **the measurement arrangement**, which captures the waveform, from **signal processing**, which extracts the fault information.

A. Direct voltage sampling—the simplest starting method

Inject a low-voltage pulse or fast voltage step, then measure the voltage at the cable's input using an oscilloscope or digitizer. This is the basic TDR arrangement: an impedance discontinuity sends part of the signal back toward the measurement point. (Keysight)

Ideally, the measured voltage contains:

$$V_{\text{measured}}(t) = V_{\text{incident}}(t) + V_{\text{reflected}}(t)$$

The practical difficulty is that **a nearby fault's echo may overlap the transmitted waveform or its settling response**. Generator imperfections, connections, and measurement fixtures can obscure the reflection, so calibration is important. (Keysight)

My recommendation: Begin with this arrangement to establish the cable's response and validate your analysis software.

B. Reflection bridge or directional coupler—separating outgoing and returning waves

A reflection bridge or suitable directional coupler provides a measurement output that preferentially responds to the returning wave. Its ability to reject the outgoing wave is associated with its **directivity**; imperfect directivity allows some transmitted signal to leak into the reflection channel. (Mini-Circuits Blog -)

For your prototype, I would consider a **broadband, time-domain-compatible reflection bridge** rather than assume that an arbitrary RF coupler is suitable. Check its lower and upper frequency limits, pulse response, insertion loss, and impedance matching.

Importantly, the output should go to a **waveform receiver**, not merely a power detector: you need the reflection's timing and sign, not just its energy.

My recommendation: This is the preferred next step when direct sampling cannot distinguish small or closely timed echoes adequately.

C. Simultaneous voltage and current measurement—useful for research

Measure voltage and current at approximately the same point, with synchronized channels. Under a simplified single-mode transmission-line model with real, approximately constant characteristic impedance Z_0, define current as positive toward the cable:

$$V = V^+ + V^-, \qquad I = \frac{V^+ - V^-}{Z_0}$$

Solving these equations gives:

$$V^+ = \frac{V + Z_0 I}{2}, \qquad V^- = \frac{V - Z_0 I}{2}$$

Thus, the reflected-voltage waveform can be **calculated from two measured signals**.

This is an analytical result under the stated model, not a guarantee for an arbitrary three-phase network. The voltage and current channels must have calibrated gain, polarity, frequency response, and relative delay. A strongly coupled multiconductor line requires a modal or matrix treatment rather than one universal scalar Z_0.

My recommendation: Use this as an advanced thesis experiment, not as the first prototype. It introduces an additional sensor and more calibration requirements.

D. Reference-trace comparison—helping identify changes

Record a healthy-line waveform, then compare it with later measurements:

$$\Delta V(t) = V_{\text{test}}(t) - V_{\text{healthy}}(t)$$

This can make new reflections easier to recognize against existing connector and junction responses. Reference or “golden device” comparisons are established TDR techniques, but the connection arrangement and operating configuration must remain sufficiently repeatable. (Tektronix)

This is **an addition to methods A–C**, not a separate sensor.

Frequency-domain reflectometry is another alternative: it measures reflections over a frequency sweep and transforms the result into a time-domain response. It does not directly capture an injected short pulse in the same way as pulse TDR. (Tektronix)

2. Which sensors are best?

First choice: wideband voltage sensing

For your offline project, I would prioritize a **low-capacitance differential voltage probe** or a **purpose-built, calibrated voltage-sampling receiver**.

A differential probe measures the voltage between the selected conductors. However, a differential probe does not automatically separate incident and reflected waves; that is the job of the bridge, timing analysis, or wave-separation processing.

The important properties are **bandwidth, input loading, noise, pulse fidelity, and short, repeatable connections**. Probe and fixture parasitics can degrade TDR measurements significantly. Also, a differential probe is not automatically galvanically isolated. (Tektronix)

Why I prefer voltage sensing: It is the most direct route to the voltage waveform used in conventional TDR interpretation.

Second choice: high-frequency current transformer—HFCT

An HFCT detects the pulse current magnetically. A suitable device can have a very fast response, but **“current transformer” alone is not a sufficient specification**. Select a pulse-current monitor whose

usable rise time is faster than the current transition you need to measure. Pearson's guidance explicitly emphasizes this requirement. (Pearson Electronics)

Check both ends of its frequency range. The upper limit affects edge fidelity, while the lower-frequency response causes **pulse droop**: a flat current pulse may appear to slope downward because a transformer cannot pass DC. (Pearson Electronics)

My recommendation: An HFCT is the best supplementary sensor for validating pulse current or implementing the V/I separation method. Do not substitute an ordinary power-frequency CT without verifying its transient performance.

Magnetic coupling also does **not** make every HFCT safe around live conductors; insulation requirements still apply. (Pearson Electronics)

Conditional choice: Rogowski coil

A Rogowski coil produces a raw output proportional to the **rate of change of current**:

$$V_{\text{coil}} \propto \frac{dI}{dt}$$

An integrator is needed to obtain a signal proportional to current. Its air-core construction avoids magnetic-core saturation, but the coil, connecting cable, and integrator all influence bandwidth and phase response. (PEM UK)

My recommendation: Useful for large-current or specialized transient measurements, but not my default choice for small reflected pulses with nanosecond edges. A particular Rogowski system may work well; it must be judged by its actual response, not by the sensor name.

Selection summary

Your objective	My preferred choice
First offline TDR prototype	Wideband voltage sensing and an oscilloscope
Better separation of outgoing pulse and returning echo	Reflection bridge and waveform receiver
Simultaneous voltage/current research	Differential voltage sensing plus HFCT
Large-current transient research	Suitable Rogowski system or pulse-current transformer

These are **engineering recommendations for your project**, rather than a claim that one sensor is universally best.

3. What bandwidth and sampling rate should you target?

Select the receiver for the pulse's rise time—not merely its width or repetition frequency.

For a first-order receiver, the mathematical relationship between its 10–90% rise time and its −3 dB bandwidth is approximately:

$$t_{r,\,\text{receiver}} \approx \frac{0.35}{B}$$

This follows from the first-order step response, for which $t_r = \ln(9)\,\tau$ and $B = 1/(2\pi\tau)$.

For example, suppose your experimental pulse has a **5 ns rise time**. A receiver whose own rise time is also 5 ns would noticeably broaden that edge. As an initial laboratory design target, I would therefore consider:

Parameter	Illustrative starting target
Receiver bandwidth	Approximately **200 MHz**
Real-time sampling rate	Approximately **1 GS/s or higher**
Channels	Two for reference/echo or voltage/current comparison
Recording	Raw waveforms, with repeatable triggering and averaging
Calibration	Known test loads, fixture delay, and cable propagation velocity

These are **proposed prototype targets**, not established requirements for your installation. At 200 MHz, the first-order receiver rise-time estimate is 1.75 ns; at 1 GS/s, samples are 1 ns apart. Neither number alone establishes distance accuracy.

Also, my earlier pulse-width discussion needs a qualification: **pulse duration affects energy and waveform overlap, while edge speed and the complete measurement response also affect what you can resolve.** Fault-location accuracy should not be equated directly with $v T_p/2$. (Keysight)

4. What changes for your three-phase parallel line?

For research, I would plan measurements across **L1–L2, L2–L3, and L3–L1**, with the other conductors' connections documented and controlled. These are proposed test configurations—not a guarantee that three traces uniquely identify every fault.

Coupling between conductors and reflections from parallel branches complicate interpretation. **A better sensor cannot, by itself, remove ambiguity caused by network topology.** (Tektronix)

One particularly important consequence of the wave equations above is:

$$I^- = -\frac{V^-}{Z_0}$$

Therefore, **do not apply a reflected-voltage polarity rule directly to an HFCT waveform** without accounting for the current direction and sensor orientation.

My final recommendation

For your master's project, start with **offline pulse TDR using wideband voltage sensing**, validate it on known open and short terminations, and then add a **calibrated reflection bridge**. Add an **HFCT channel** when you are ready to investigate current-wave behavior and voltage/current wave separation.

That sequence gives you a simpler baseline first and a meaningful, testable research extension afterward.

09 · User — First download request

can i download all of these chat?

10 · User — Repeated download request

can i download all of these chat?

Archive appendix · Source-link directory

These are the web addresses linked in the exported replies, not a newly verified bibliography. Tracking parameters are omitted from the displayed directory addresses; the original hyperlinks are preserved in the transcript. Source URLs behind the first answer's chat-only citation markers were not available.

1. **keysight.com** — https://www.keysight.com/us/en/assets/7018-01451/application-notes/5989-5723.pdf
2. **osha.gov** — https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.333
3. **tek.com** — https://www.tek.com/en/support/faqs/what-causes-reflections-tdr
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9. **pearsonelectronics.com** — https://www.pearsonelectronics.com/
10. **pearsonelectronics.com** — https://www.pearsonelectronics.com/support/application-notes/specifications
11. **pemuk.com** — https://www.pemuk.com/support/the-rogowski-coil
12. **tek.com** — https://www.tek.com/en/support/faqs/how-bandwidth-related-rise-time-oscilloscopes
13. **Mini-Circuits Blog -** — https://blog.minicircuits.com/directional-couplers-their-operation-and-application/
14. **Pearson Electronics** — https://www.pearsonelectronics.com/support/application-notes/rise-time
15. **Pearson Electronics** — https://www.pearsonelectronics.com/support/application-notes/droop-notes
16. **Pearson Electronics** — https://www.pearsonelectronics.com/support/application-notes/high-voltage-consideration
17. **PEM UK** — https://www.pemuk.com/how-it-works.aspx